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Membrane proteins involved in potassium shifts during muscle activity and fatigue

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Kristensen, Michael, Thomas Hansen, and Carsten Juel. Membrane proteins involved in potassium shifts during muscle activity and fatigue. *Am J Physiol Regul Integr Comp Physiol* 290: R766–R772, 2006. First published October 13, 2005; doi:10.1152/ajpregu.00534.2004.—Muscle activity is associated with potassium displacements, which may cause fatigue. It was reported previously that the density of the large-conductance Ca^{2+} -dependent K^+ (BK_{Ca}) channel is higher in the T tubule membrane than in the sarcolemmal membrane and that the opposite is the case for the ATP-sensitive K^+ (K_{ATP}) channel. In the present experiments, we investigated the subcellular localizations of the strong inward rectifier 2.1 K^+ (Kir2.1) channel and the Na^+ - K^+ - 2Cl^- (NKCC1) cotransporter with Western blot analysis of different muscle fractions. Furthermore, muscle function was studied while trying to manipulate the opening probability or transport capacity of these proteins during electrical stimulation of isolated soleus muscles. All experiments were made with excised muscle from male Wistar rats. Kir2.1 channels were almost undetectable in the sarcolemmal membrane but present in the T tubule membrane, whereas NKCC1 cotransporters were present in the sarcolemmal membrane. For muscles incubated in a buffer containing pinacidil, NS1619, Ba^{2+} , or bumetanide, there was a faster reduction in peak force ($P < 0.05$). Furthermore, bumetanide incubation reduced the peak force at the onset of electrical stimulation ($P < 0.05$). Thus the effects on muscle force indicate that these drugs can affect K^+ -transporting proteins and thereby influence K^+ accumulation, especially in the T tubules, suggesting that K_{ATP} and BK_{Ca} channels are responsible for K^+ release and decrease in force during repeated muscle contractions, whereas Kir2.1 and NKCC1 may have a role in K^+ reuptake.

channels and cotransporters; T tubule

K^+ DISPLACEMENT in skeletal muscle is important for the development of muscle fatigue during intense exercise. In vitro studies have shown that an increase in the extracellular K^+ concentration ($[\text{K}^+]$) to values >10 mM significantly reduces the development of force in skeletal muscle (4, 5, 24, 38). This is probably because of a depolarization of the muscle fiber, followed by an impairment of action potential (AP) propagation along the sarcolemmal and T tubule membrane. It has been shown in several human studies in vivo that interstitial $[\text{K}^+]$ increases progressively with increasing work intensity, reaching values of 9–10 mM at exhaustion (26, 36). The extracellular increase in $[\text{K}^+]$ is expected to be even more pronounced in the T tubule than in the interstitium, because of the relatively long and narrow shape of the T tubule and the absence of blood flow (35, 37). K^+ homeostasis in the T tubule is consequently expected to be particularly important in the regulation of muscle fatigue.

Many membrane proteins are involved or have been proposed to be involved in the regulation of K^+ homeostasis in skeletal muscle both at rest and during muscle contraction. K^+ ions are released from skeletal muscle through the voltage-sensitive K^+ (K_v) channels terminating the APs (21, 43), the large-conductance Ca^{2+} -dependent K^+ (BK_{Ca}) channels (23, 31), and the ATP-sensitive K^+ (K_{ATP}) channels (29, 30, 36). The reuptake of K^+ is mainly performed by Na^+ - K^+ -ATPase (for a review see Ref. 6), but the Na^+ - K^+ - 2Cl^- (NKCC1) cotransporter is also involved (30, 48). Moreover, strong inward rectifier channels might participate in K^+ reuptake during muscle contraction (21, 43, 46) and account for the inward rectification current measured in skeletal muscle (7, 22, 44). In this regard, one candidate is the strong inward rectifier 2.1 K^+ (Kir2.1) channel, which is present in skeletal muscle (3). Whereas the subcellular distribution of the K_{ATP} and BK_{Ca} channels between the T tubule and sarcolemmal membrane has already been investigated (36), the subcellular localizations of both the NKCC1 cotransporter and the Kir2.1 channel in skeletal muscle are unknown. When studying K^+ -regulating proteins in skeletal muscle, it is obvious to look at the different fiber types, as oxidative and glycolytic fibers show marked differences in the development of muscle fatigue. This divergence could be caused by differences in the amounts/densities of the K^+ -regulating membrane proteins in different fiber types.

The opening probability (P_o) of channels and the transport capacity (T_c) of a cotransporter could be other important factors in determining K^+ displacement/homeostasis in skeletal muscles. If P_o or T_c values are relatively low during muscle contraction, the transport system is expected to contribute very little to K^+ regulation. The effect of manipulating the P_o of the K_{ATP} channels, both at rest and during muscle contraction, has been thoroughly investigated (17, 29, 30, 32, 36), but less is known about the other K^+ -regulating proteins.

The aims of this study were 1) to investigate the subcellular distribution of the Kir2.1 channel and the NKCC1 cotransporter, 2) to characterize their fiber type distribution, 3) to compare the subcellular distribution of the Kir2.1 channel and NKCC1 cotransporter with the subcellular distribution of K_{ATP} and BK_{Ca} channels, and 4) by manipulating their P_o or T_c , to test whether the Kir2.1 channel, the BK_{Ca} channel, and the NKCC1 cotransporter are important at rest and during muscle contraction, respectively.

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MATERIALS AND METHODS

Western blot analysis was used to investigate the subcellular localization and fiber-type distribution of the investigated proteins, and electrical stimulation was used in an attempt to study their physiological effects. Rats used for both Western blot analysis and electrical stimulation were killed by cervical dislocation; *n* indicates the number of animals. The handling of animals was in accordance with Danish Animal Welfare Regulations.

Western blot analysis. Three different muscle fractions were prepared from male Wistar rats (90–100 g). One set of muscle was excised for production of sarcolemmal giant vesicle fractions (25), and another set was excised to make two different kinds of muscle homogenate fractions (hom1 and hom2; Fig. 1). Furthermore, fiber type-specific sarcolemmal giant vesicle and homogenate fractions (including both hom1 and hom2) were prepared to investigate whether there were differences in fiber type distributions of the investigated proteins. Fiber type-specific samples containing glycolytic or oxidative muscle fibers were made from the following muscles: white gastrocnemius, white vastus lateralis, and white tibialis anterior (glycolytic samples) and soleus, red gastrocnemius, and vastus intermedius (oxidative samples). The fiber-type distribution (I:IIA:IID/X:IIB) in these preparations has been calculated according to Delp and Duan (11) to be ~0:0:4:96 (glycolytic) and 69:23:7:1 (oxidative).

The dihydropyridine (DHP) receptor was used as a membrane and fiber-type marker. This receptor is exclusively located in the T tubule membrane (15) and is therefore a good indicator for the presence of T tubule membrane in the samples. The sarcolemmal giant vesicle fractions do not contain any DHP receptors (36) and therefore represent purified sarcolemmal membrane. Fiber-type specificity can be evaluated because of the three to five times higher number of DHP receptors found in glycolytic compared with oxidative skeletal muscle fibers (Refs. 10 and 28; Fig. 1).

Sarcolemmal giant vesicle fractions were prepared according to Juel (25). For the homogenate fractions, muscles were homogenized in sucrose buffer (in mM: 250 sucrose, 30 HEPES, 2 EGTA, 40 NaCl, 2 PMSF, pH 7.4) with a Polytron 2100 homogenizer. After homogenization the samples were centrifuged at 1,000 *g* for 1 min, and the pellet (designated hom2) was resuspended in Tris-SDS (10 mM Tris, 4% SDS, 1 mM EDTA, 2 mM PMSF, pH 7.4). The supernatant was spun in a high-speed ultracentrifuge (190,000 *g*) for 90 min at 4°C. The resulting pellet (designated hom1) was then resuspended in Tris-SDS. Both hom1 and hom2 are homogenate samples, but they contain different ratios of membrane proteins compared with cellular proteins. Protein concentrations in the sarcolemmal giant vesicles and homogenates were measured with a BSA standard (DC protein assay,

Bio-Rad) and analyzed on a ThermoSpectronis Genesys 10vis (VWR). The samples were diluted with Tris-SDS until all samples contained the same concentration of protein and then mixed 1:1 with sample buffer (10% SDS, 5% glycerol, 10 mM Tris·HCl, 1 mM EDTA, 10 mM DTT). The proteins were separated on SDS-PAGE (Excell 8–18% gradient gel) and thereafter electroblotted for 1 h onto a Millipore Immobilon-P polyvinylidene difluoride membrane. The membrane was blocked for 1 h in a buffer containing 2% BSA with 1% low-fat dry milk (5% low-fat dry milk and no BSA for the membrane for NKCC) and 0.1% Tween 20. The primary and secondary antibodies were diluted in the same buffer solution (same amount of low-fat dry milk, BSA, and Tween 20) as used for the membrane blocking. Monoclonal anti-DHP-receptor α_1 -subunit antibodies were obtained from Affinity Bioreagents (catalog no. 3-920). Both the Kir2.1 subunit antibody (catalog no. AB5374) and the NKCC1 subunit antibody (catalog no. AB3560P) were obtained from Chemicon (both polyclonal antibodies). The membrane was incubated overnight with the primary antibody and thereafter for 1.5–2 h at room temperature with the secondary antibody. After repeated washing in Millipore H₂O with 0.05% Tween 20 and 1 M NaCl (only Millipore H₂O for the last wash), the membrane was incubated with enhanced chemiluminescence (ECL)– or ECL+ Western blotting detection reagents (Amersham Biosciences) at room temperature and visualized on hyperfilm ECL (Amersham Biosciences; Fig. 1). The hyperfilm was scanned, and the intensities of the bands (assumed to be proportional to protein amount in the sample) were analyzed with SigmaGel software version 1.0.

The amount of the Kir2.1 and NKCC1 proteins in the membrane were compared with the corresponding amounts of the DHP receptor measured in the same fractions (Fig. 1). The membranes were first used to investigate the amount of Kir2.1 channels or NKCC1 cotransporters in two or three samples from each of the three muscle fractions (vesicles, hom1, and hom2). The membranes were stripped (Re-Blot Plus; catalog no. 2501, Chemicon) and reused to determine the amount of the DHP receptor. Because of the distribution of the Kir2.1 channel and NKCC1 cotransporter (see RESULTS), homogenate samples were used for study of fiber-type dependence for the Kir2.1 channel, whereas sarcolemmal giant vesicle samples were used for the NKCC1 cotransporter. Samples were always compared from the same gel and with identical amounts of protein per lane. The presence of the DHP receptor was quantified from the amount of the α_1 -subunit (200 kDa). The presence of the Kir2.1 channel was quantified from the amount of the Kir2.1 subunit (~60 kDa). Bands of approximately similar size (55–60 kDa) have been measured in other tissues (34, 49). The presence of the NKCC1 cotransporter was demonstrated by the 12-transmembrane subunit, which showed a band at ~170 kDa. Bands at similar sizes were seen by Alvarez et al. (1).

Statistics for Western blot analysis. For the subcellular distribution, we used Student's paired *t*-tests to investigate whether there are differences between the amount of DHP receptor, Kir2.1 channel, and NKCC1 cotransporter in the two homogenate fractions (hom1 and hom2). This analysis was not carried out for the vesicles because the sample size was too small and because the purification of the vesicles is well established (36). Student's paired *t*-tests were also used to compare the amounts of the investigated proteins in the different fiber types. *P* values <0.05 were considered statistically significant.

Electrical stimulation. Both soleus muscles were isolated from male Wistar rats (65–85 g) immediately after they were killed. The muscles were placed randomly in two different baths of Krebs-Ringer solution (in mM: 122 NaCl, 25 NaHCO₃, 2.8 KCl, 1.2 KH₂PO₄, 1.2 MgSO₄, 1.3 CaCl₂, 5.0 D-glucose), one as control and the other as treatment. The names, concentrations, and functions of the different drugs used to manipulate *P*_o and *T*_c for the different proteins (channels and cotransporter) are listed in Table 1. It should be noted that Ba²⁺ is a nonspecific channel blocker at high concentrations. However, at low concentrations (<100 μ M) Ba²⁺ has been reported to specifically inhibit Kir2.1 (42, 47). When the drugs were initially dissolved in

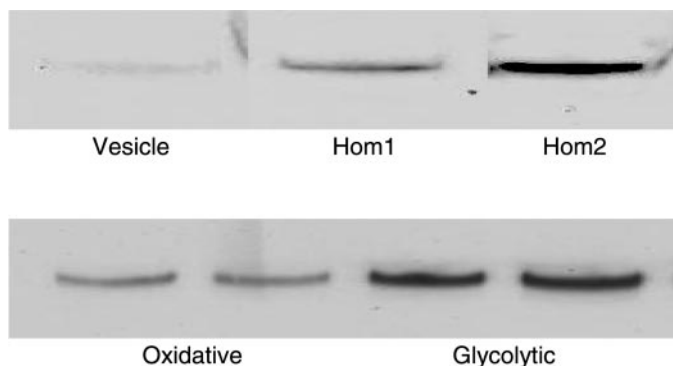


Fig. 1. Dihydropyridine (DHP) receptor bands in the different fractions. *Top*: an example of the amount of the DHP receptor in the 3 different fractions (vesicles, hom1, and hom2) used to investigate the localization of the Kir2.1 channel and Na⁺-K⁺-2Cl⁻ (NKCC)1 cotransporter. *Bottom*: examples of the amounts of DHP receptors in oxidative and glycolytic samples (including both hom1 and hom2) used to investigate fiber-type differences.

Table 1. *Drugs used during electrical stimulation of skeletal muscle*

Drug	Function	Concentration, μM	References	Distributor
BaCl ₂	Kir2.1 channel inhibitor	5	7, 42, 47	Sigma-Aldrich (S05472-531)
Pinacidil	K _{ATP} channel opener	100	32	Sigma-Aldrich (P154)
NS-1619	BK _{Ca} channel opener	35	39	Sigma-Aldrich (N170)
Recombinant slotoxin	BK _{Ca} channel inhibitor	0.050	16	Alomone Labs (RTS-410)
Recombinant iberiotoxin	BK _{Ca} channel inhibitor	0.050	33	Alomone Labs (RTI-400)
Bumetanide	NKCC inhibitor	100	48	Sigma-Aldrich (B-3023)

The references cited are the most important guidelines for choosing the different concentrations during the procedure. However, there is some divergence, mainly because of differences in animal size, incubation temperature, and time. K_{ATP}, ATP-sensitive K⁺; BK_{Ca}, large-conductance Ca²⁺-dependent K⁺; NKCC, Na⁺-K⁺-2Cl⁻ cotransporter.

DMSO, the same amount of DMSO was added to the respective control muscle during incubation. Krebs-Ringer solution was equilibrated with a mixture of 5% O₂-95% CO₂ at room temperature before and during the experiment (pH 7.4) and kept at ~26°C during the procedure.

After 1 h of incubation in either Krebs-Ringer solution (control) or Krebs-Ringer solution with the dissolved drug (treated), the muscles were fixed to a force transducer and adjusted to produce the same passive force. Ag⁺ electrodes (separated ~3 mm) were placed on each side of the muscle. The muscles were stimulated for 1 s every 3 s (pulse duration 2 ms, electrical field $\approx$ 100 V/cm) with a frequency of 33 Hz over 6 min. The developed force was recorded by an analog-to-digital converter (Duo-18 version 1.1; data for analyzing were read every 30 s). Forces measured during electrical stimulation are expressed as a percentage of the initial force measured at the start of the experiments ($t = 0$). Results from the K_{ATP} channel experiment (see RESULTS) were published previously as an example to explain the use of different statistical models (27).

Statistics for electrical stimulation. Data from each of the five separate experiments were analyzed with a two-way repeated-measures ANOVA. Usually, this model includes three fixed effects: time, treatment, and the interaction between time and treatment. Normally, however, one would not be interested in either the treatment effect or the interaction effect per se (27). Instead the most interesting—and easily interpretable—hypothesis would be to test whether the 12 pairwise differences between the 2 treatments (1 for each time point measurement) would all be the same (i.e., zero). This effect is in fact the interaction between time and treatment when controlled for time only and, accordingly, the aforementioned hypothesis is tested comparing the two nested models

$$Y_{ij} = \text{time}_i + \text{time} \cdot \text{treatment}_{ij} + e_{ij}$$

with

$$Y_{ij} = \text{time}_i + \text{treatment}_{ij} + e_{ij}$$

where subscripts $i = 1, 2$ refer to the 2 treatments, subscripts $t = 1, 2, \dots, 12$ are the 12 different time points, and $j = 1, 2, \dots$ is the animal indicator, using the likelihood ratio test with 11 degrees of freedom. Here, Y_{ij} denotes the relative force of the j th animal from the i th group to time t , while e denotes the residual or unexplained error. Student's paired t -tests were used to compare forces between the muscles at $t = 0$. Analyses were performed with SAS Proc mixed. P values < 0.05 were considered statistically significant.

RESULTS

Distribution of Kir2.1 channel and NKCC1 cotransporter. Both the Kir2.1 channel and the DHP receptor were absent in the sarcolemmal giant vesicle fractions but present in both homogenate fractions (hom1 and hom2) (Fig. 2A). The opposite result was found for the NKCC1 cotransporter, in that the amount of the NKCC1 cotransporter was higher in the sar-

colemmal giant vesicle fraction than in both homogenate fractions (Fig. 2B).

Fiber-type dependence. The relative amount of the DHP receptor was 2.9 times higher [95% confidence interval (CI): 2.0–4.2] in homogenate samples made from glycolytic fibers than in samples made from oxidative fibers (Fig. 3; $P < 0.05$, $n = 4$). For the Kir2.1 subunit the relative amount was 7.9 times higher (95% CI: 5.8–10.9) in samples made from glycolytic fibers (homogenate samples) than in samples made from oxidative fibers (Fig. 3; $P < 0.05$, $n = 5$), whereas the relative amount of the NKCC1 cotransporter was 1.80 times higher (95% CI: 1.4–2.3) in the glycolytic fiber samples than in the oxidative fiber samples (Fig. 3; $P < 0.05$, $n = 3$).

Manipulation of P_o and T_c during electrical stimulation. For muscles incubated in a buffer containing 100 μM pinacidil (an expected K_{ATP} channel opener), the development of peak force

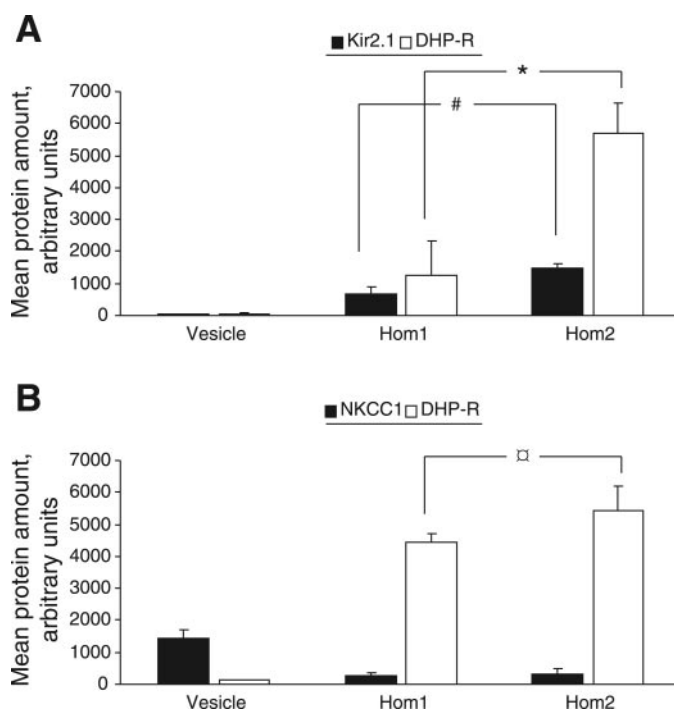


Fig. 2. Protein amounts of DHP receptor and Kir2.1 channel (A) and NKCC1 cotransporter (B) in sarcolemmal giant vesicle fractions and homogenate fractions (hom1 and hom2). Values are means $\pm$ SD ($n = 2$ in vesicles; $n = 3$ in hom1 and hom2). Significant difference between the amounts of Kir2.1 channel in hom1 and hom2: #Kir2.1, $P = 0.0458$; *DHP-R, $P = 0.0101$ (A); significant difference between the amounts of DHP receptor in hom1 and hom2: □ $P = 0.0414$ (B).

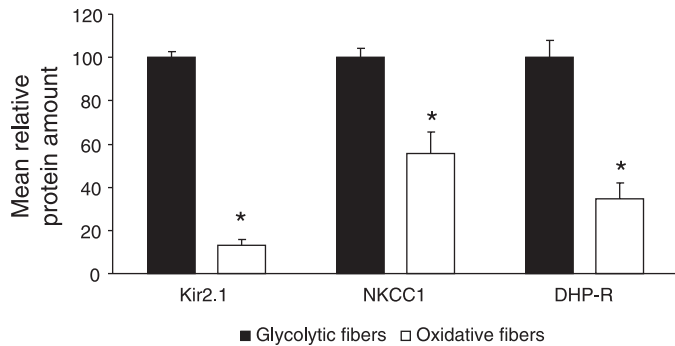


Fig. 3. Relative protein amounts of Kir2.1 channel, NKCC1 cotransporter, and DHP receptor (means $\pm$ SD) in samples containing either glycolytic or oxidative muscle fibers. Kir2.1 channel and DHP receptor were investigated in homogenate fractions (including both hom1 and hom2), whereas NKCC1 cotransporter was investigated in sarcolemmal giant vesicle fractions. The measured mean protein amounts for glycolytic fibers in each pair (glycolytic and oxidative) have been multiplied with a given factor to reach 100. The oxidative fibers and the SD have been multiplied with the same factor. *Relative amounts for the measured proteins are significantly higher in the glycolytic fibers compared with the oxidative fibers: Kir2.1, $P < 0.0001$; NKCC, $P = 0.0085$; and DHP-R, $P = 0.0022$.

decreased significantly faster during electrical stimulation compared with control muscles (Fig. 4A; $P < 0.05$, $n = 7$). In addition, there was a faster decrease in the development of peak force ($P < 0.05$, $n = 10$) for muscles incubated in a buffer containing 40 μ M NS-1619 (expected BK_{Ca} channel opener) than in control muscles (Fig. 4B). On the other hand, there was no difference in the decrease of peak force for muscles incubated in a buffer with 50 nM recombinant iberiotoxin and recombinant slotoxin (expected BK_{Ca} channel inhibitors) compared with control muscles (Fig. 4C; $P > 0.05$, $n = 8$). For muscles incubated in a buffer containing 5 μ M Ba²⁺ (expected Kir2.1 channel inhibitor), the development of peak force decreased significantly faster during electrical stimulation than for control muscles (Fig. 4D; $P < 0.05$; $n = 7$).

In addition, there was a significantly faster decrease in peak force for muscles incubated in a buffer containing 100 μ M bumetanide (an expected NKCC inhibitor) (Fig. 4E; $P < 0.05$, $n = 8$) compared with control muscles. In the bumetanide experiment there was a significant difference in the development of peak force at $t = 0$ between treated and control muscles (Fig. 5; $n = 9$) but not for any of the other groups (Fig. 5; $n = 7$ –10).

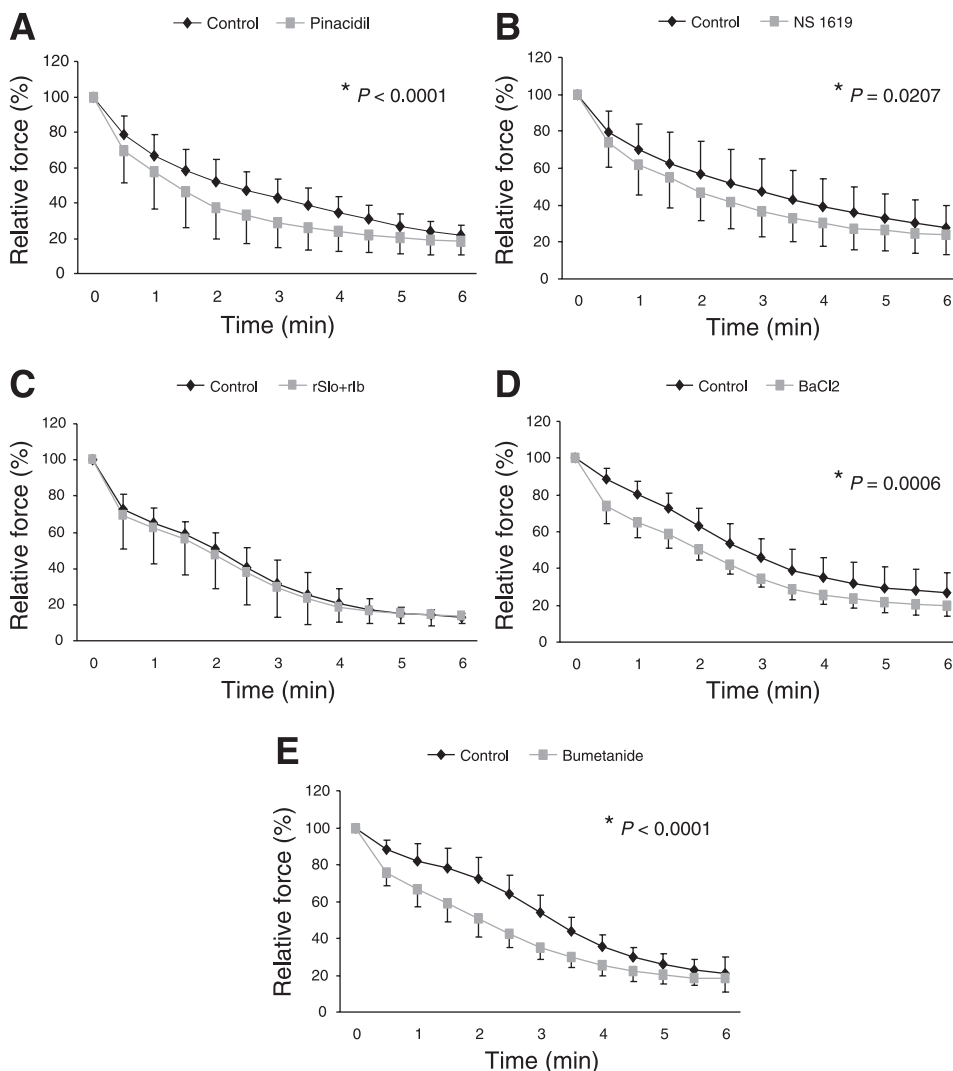


Fig. 4. Development of force during continuous stimulation of control muscle and muscle incubated with stimulators or inhibitors of the cotransporter and different K⁺ channels (for concentrations and functions of the drugs, see Table 1). A: pinacidil. B: NS-1619. C: recombinant slotoxin (rSlo) + recombinant iberiotoxin (rIb). D: Ba²⁺. E: bumetanide. Average $\pm$ SD peak force for all muscles in each treatment group is shown. Force was measured relative to the force at the start of the experiments ($t = 0$) for each muscle. The muscles were stimulated for 6 min with 1-s trains (33-Hz pulse) with a break of 2 s between each train. To simplify the figure, data are shown for every 30 s. *Average development of peak force for the threaded muscles differed significantly from that for the control muscles.

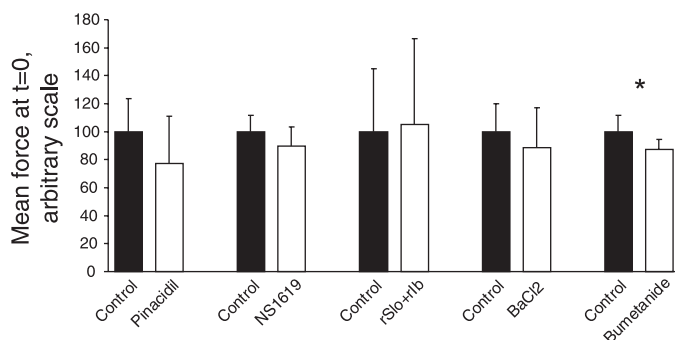


Fig. 5. Mean force at $t = 0$. Different stimulators and inhibitors of the cotransporter and K^+ channels were tested after 1 h of incubation. There was a significant difference in the development of peak force at $t = 0$ for bumetanide-treated muscles ($n = 8$). None of the other groups (treated vs. control) showed any significant difference ($n = 7$ – 10). The mean $\pm$ SD force in control muscles was set to 100 by multiplying with a given factor (arbitrary units). The absolute force for the different muscles lies between 0.25 and 0.30 N. For incubation concentrations and function for the drugs used see Table 1.

DISCUSSION

Adding pinacidil (a K_{ATP} channel opener), NS1619 (a BK_{Ca} channel opener), Ba^{2+} (a $Kir2.1$ channel inhibitor at low concentrations; Refs. 42 and 47), or bumetanide (a NKCC1 cotransporter inhibitor) to the incubation medium all significantly increased the rate of peak force decline during electrical stimulation (Fig. 4A). For pinacidil this is in accordance with Matar et al. (32). Except for bumetanide, none of the drugs had an effect on the development of peak force at $t = 0$ (Fig. 5). For pinacidil, this is also in agreement with Matar et al. (32). Furthermore, Gong et al. (18) showed that there was no effect on the resting membrane potential, half-relaxation time, or width of single action potentials when increasing K_{ATP} channel P_o . Even though there were no effects of adding recombinant iberiotoxin and recombinant slo toxin (BK_{Ca} channel inhibitors) to the incubation medium (Fig. 4C), the present findings support the idea that K^+ plays an important role in muscle fatigue and indicate that both the K_{ATP} , BK_{Ca} , and $Kir2.1$ channels and the NKCC1 cotransporter have an important function.

The absence of $Kir2.1$ in the sarcolemmal giant vesicles (Fig. 2) suggests that $Kir2.1$ channels are predominantly located in the T tubule membrane of skeletal muscle. This is further supported by the fiber-type differences. The relative amount of the $Kir2.1$ channel is significantly higher in glycolytic muscles compared with oxidative muscles (Fig. 3). This is probably due to higher amounts of T tubules in glycolytic muscles (10, 28) and/or a higher protein density. This localization therefore suggests that the $Kir2.1$ channel has an important role in maintaining K^+ homeostasis in the T tubule, and perhaps in determining the onset of muscle fatigue during intense exercise (see introduction). The NKCC1 cotransporter is present in the sarcolemmal membrane (Fig. 2); however, it cannot be determined whether, and to what extent, it is present in the T tubule membrane. Furthermore, the relative amount of the NKCC1 cotransporter in the sarcolemmal membrane of glycolytic fibers was found to be almost twice that in oxidative fibers. Higher ion displacements occur in glycolytic muscle fibers during muscle contraction (36). The demand for transport proteins to rebuild the ion gradients (and water balance) is

therefore greater, which is in line with the higher amount of NKCC1 cotransporter in the glycolytic muscle fiber sarcolemmal membrane.

Perspectives

Overall, our experiments attempting to influence the time to fatigue when manipulating P_o for K_{ATP} and BK_{Ca} channels suggest that they are both involved in K^+ release during muscle contraction. This is supported by the different stimuli that affect the P_o for K_{ATP} and BK_{Ca} channels. Decreases in ATP (ATP-to-ADP ratio; Ref. 45) and pH (9) have been shown to increase P_o for the K_{ATP} channel, whereas increased Ca^{2+} concentration (23) and stretch of the cell membrane (31) have been shown to increase P_o for the BK_{Ca} channel. These stimuli increase during muscle contraction and in vivo might increase P_o , resulting in a higher K^+ release and thereby a depolarization of the muscle fiber. In addition, the increased K_{ATP} channel activity might counteract Na^+ -dependent depolarization during AP (18).

The present stimulation experiments were carried out in vitro and were therefore without blood flow. It is therefore possible that all the K^+ released across the sarcolemmal membrane, and not transported back by either Na^+ - K^+ -ATPase or the NKCC1 cotransporter, accumulates in the interstitium and that extracellular K^+ accumulates faster and to a higher concentration than in vivo. This problem is expected to be less pronounced in the T tubule because only a small fraction of the K^+ from the T tubules is expected to diffuse into the interstitium and be removed by the blood (2, 35, 37). The K_{ATP} channel is opposite the BK_{Ca} channel predominantly located in the sarcolemmal membrane (36). The effect of increasing K_{ATP} channel P_o might therefore partly be caused by an artificially increased $[K^+]$ in the interstitium, rather than an increased $[K^+]$ in the T tubule. The inhibitors recombinant iberiotoxin and recombinant slo toxin had no effect when added to the incubation medium. A possible explanation could be that the diffusion rates for these drugs were lower than for NS-1619 (the inhibitors are >10 times bigger than the opener) and could therefore be dependent on blood flow to reach the BK_{Ca} channels. In situ experiments or increasing the temperature to 37°C may overcome these problems. Finally, blocking the BK_{Ca} channel could possibly result in a greater activation of other channels (e.g., the K_{ATP} channel) so that the effect is reduced. Whether this happens can be investigated either by direct measurements of K^+ (or Rb^+) fluxes or by inhibition of both channels during electrical stimulation.

The NKCC cotransporter mediates an electroneutral ion transport in skeletal muscle under normal physiological ion concentrations. In resting skeletal muscles, it accounts for $\sim 15\%$ of the K^+ uptake (30) and $\sim 23\%$ of the Na^+ uptake (12), and it has been shown to account for $>30\%$ of the K^+ transport during muscle stimulation with either catecholamines or electrical stimulation (19, 48). Our results also suggest that the NKCC1 cotransporter has an effect on K^+ homeostasis, both at rest and during muscle contraction. The effect of reducing T_c for the NKCC1 cotransporter could influence K^+ regulation in two ways. One way is a direct reduction of the K^+ reuptake during muscle contraction as shown by Wong et al. (48), which could result in a faster increase in the extracellular $[K^+]$ and speed up time to fatigue. Another way could be to

decrease $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity by reducing Na^+ influx. Increasing $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity has been shown to protect against K^+ -induced muscle fatigue (35, 40), probably by reducing extracellular K^+ . A reduction of $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity is expected to have the opposite effect. To investigate this possibility, it is necessary to directly measure K^+ (or Rb^+) fluxes in the muscle. Finally, one must remember that the lack of blood flow can influence the result. During muscle contraction the concentrations of Na^+ , K^+ , and/or Cl^- in the interstitium could reach values different from those seen in vivo. This could influence the T_c for the NKCC1 cotransporter. Again, in situ experiments are necessary to avoid this problem.

Inward K^+ currents have been measured in skeletal muscle (7, 22, 44). The Kir2.1 channel, which shows strong inward rectifying properties (20, 14), could be responsible for a part of this current. During muscle contraction, this current has been suggested to mainly take place across the T tubule membrane (46). If this is correct, the rising $[\text{K}^+]$ in the T tubule causes the equilibrium potential for K^+ to become positive relative to the resting membrane potential (because a high Cl^- conductance will try to keep the resting membrane potential stable; Refs. 8 and 13), which will reverse the driving force for K^+ (8, 21, 46). Wallinga et al. (46) concluded that if the driving force for K^+ changes in the T tubule during intense exercise, the function of the inward rectifier channels could be to delay the onset of muscle fatigue. This is in line with our finding that Ba^{2+} , which both reduces the inward K^+ current (7, 44) and blocks the Kir2.1 channel (41) in a time- and dose-dependent manner, induced a faster reduction in the development of force. If it occurs, the K^+ reuptake through the Kir2.1 channel probably takes place between APs. This is because the channel P_o decreases during depolarization (21) and because the driving force for K^+ only can be reversed between APs. This is in contrast to the BK_{Ca} channels, which behave much like the K_v channels and are expected to release K^+ during APs (21).

Numerous experiments have investigated the effect of K^+ displacement during muscle contraction, but except for $\text{Na}^+\text{-K}^+\text{-ATPase}$ and K_{ATP} channels, which have been shown to be important regulators of K^+ homeostasis in skeletal muscle, only a few other proteins involved in this K^+ displacement have been investigated. Our results indicate that BK_{Ca} , the Kir2.1 channels, and the NKCC1 cotransporter also play an important role in the K^+ regulation of skeletal muscle both at rest and during muscle contraction. These results lead to the suggestion that K_{ATP} and BK_{Ca} channels participate significantly in the K^+ release and the following decrease in force during fatigue, whereas the Kir2.1 channel and NKCC1 cotransporter may participate in K^+ reuptake and thereby delay muscle fatigue.

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